



Topic B14

Drugs

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B15.1	Drugs and medicinal use
B15.2	Antibiotics
B15.3	Heroin and alcohol
B15.4	Tobacco smoking

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Medicinal drugs

A drug is a substance taken into the body that modifies chemical reactions. Medicinal drugs prevent or treat disease or relieve symptoms and should be tested for effectiveness, safe dose and side effects. Placebo-controlled, double-blind trials reduce bias.

Examiner insight

“Natural” does not automatically mean safe; effect depends on dose and biological action.

2. Antibiotics

Antibiotics kill bacteria or inhibit their growth but do not treat viral infections. Random mutations create variation in bacterial populations; antibiotic exposure kills susceptible cells while resistant bacteria survive, reproduce and spread resistance. Misuse increases selection pressure.

Examiner insight

Individuals do not become antibiotic-resistant; bacterial populations evolve resistance.

3. Depressants and addiction

Alcohol slows nervous-system activity, impairs judgement and coordination, and long-term misuse can damage liver and brain. Heroin is a powerful depressant and analgesic that can cause tolerance, dependence and dangerous respiratory suppression. Addiction involves compulsive use despite harm.

Examiner insight

Tolerance means a larger dose is needed for the same effect; dependence produces withdrawal when use stops.

4. Smoking

Nicotine is addictive and raises heart rate; tar contains carcinogens and damages cilia and alveoli; carbon monoxide reduces oxygen transport. Smoking increases risks of lung cancer, chronic obstructive lung disease and coronary heart disease.

Examiner insight

Link each smoke component to a distinct mechanism and outcome.

Worked application

Situation: A person stops an addictive depressant suddenly.

Prediction: Withdrawal symptoms may occur.

Reason: Dependence has altered normal nervous-system function and the drug is no longer present.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.



Practice Paper

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Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines drug? [1]
 - A a substance taken into the body that modifies chemical reactions
 - B a medicine that kills bacteria or inhibits bacterial growth
 - C needing a larger dose for the same effect
 - D a state in which stopping a drug causes withdrawal
2. Which statement best defines antibiotic? [1]
 - A needing a larger dose for the same effect
 - B a medicine that kills bacteria or inhibits bacterial growth
 - C a state in which stopping a drug causes withdrawal
 - D an inactive treatment used as a comparison
3. Which statement best defines tolerance? [1]
 - A a state in which stopping a drug causes withdrawal
 - B an inactive treatment used as a comparison
 - C needing a larger dose for the same effect
 - D a substance taken into the body that modifies chemical reactions
4. Which statement best defines dependence? [1]
 - A an inactive treatment used as a comparison
 - B a substance taken into the body that modifies chemical reactions
 - C a medicine that kills bacteria or inhibits bacterial growth
 - D a state in which stopping a drug causes withdrawal
5. Which statement best defines placebo? [1]
 - A an inactive treatment used as a comparison
 - B a substance taken into the body that modifies chemical reactions
 - C a medicine that kills bacteria or inhibits bacterial growth
 - D needing a larger dose for the same effect
6. Which statement about medicinal drugs is correct? [1]
 - A Only an illegal substance.
 - B A drug is a substance taken into the body that modifies chemical reactions.
 - C A medicine that kills viruses.
 - D An immediate immune response.
7. Which statement about antibiotics is correct? [1]
 - A A medicine that kills viruses.
 - B An immediate immune response.
 - C Antibiotics kill bacteria or inhibit their growth but do not treat viral infections.
 - D The ability to avoid all side effects.

8. Which statement about depressants and addiction is correct? [1]
- A An immediate immune response.
 - B The ability to avoid all side effects.
 - C The highest safe drug dose.
 - D Alcohol slows nervous-system activity, impairs judgement and coordination, and long-term misuse can damage liver and brain.
9. Which statement about smoking is correct? [1]
- A Nicotine is addictive and raises heart rate; tar contains carcinogens and damages cilia and alveoli; carbon monoxide reduces oxygen transport.
 - B The ability to avoid all side effects.
 - C The highest safe drug dose.
 - D Only an illegal substance.
10. Which prediction is most appropriate when a person stops an addictive depressant suddenly? [1]
- A Only an illegal substance.
 - B An immediate immune response.
 - C Withdrawal symptoms may occur.
 - D The highest safe drug dose.

Section B · Structured questions

11 Biologists use precise vocabulary when studying drugs.

(a) Define drug. [2]

(b) Define antibiotic. [2]

(c) Distinguish tolerance from the incorrect idea: “an immediate immune response”. [2]

(d) State two important features of medicinal drugs. [2]

[Total: 8]

12 A person stops an addictive depressant suddenly.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records percentage resistant under three conditions.

condition	percentage resistant
before treatment	2
after one course	18
after repeated misuse	64

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to test an antibacterial substance using agar plates.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of drugs to an unfamiliar context.

(a) Define dependence. [2]

(b) Explain the biological importance of depressants and addiction. [3]

(c) Correct this statement: “the highest safe drug dose”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]



Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A substance taken into the body that modifies chemical reactions.
- 2 B. A medicine that kills bacteria or inhibits bacterial growth.
- 3 C. Needing a larger dose for the same effect.
- 4 D. A state in which stopping a drug causes withdrawal.
- 5 A. An inactive treatment used as a comparison.
- 6 B. “Natural” does not automatically mean safe; effect depends on dose and biological action.
- 7 C. Individuals do not become antibiotic-resistant; bacterial populations evolve resistance.
- 8 D. Tolerance means a larger dose is needed for the same effect; dependence produces withdrawal when use stops.
- 9 A. Link each smoke component to a distinct mechanism and outcome.
- 10 C. Dependence has altered normal nervous-system function and the drug is no longer present.

Solutions · Structured questions

11. (a) a substance taken into the body that modifies chemical reactions [2]. (b) a medicine that kills bacteria or inhibits bacterial growth [2]. (c) needing a larger dose for the same effect; not an immediate immune response [2]. (d) Credit any two accurate features from the guide: A drug is a substance taken into the body that modifies chemical reactions. Medicinal drugs prevent or treat disease or relieve symptoms and should be tested for effectiveness, safe dose and side effects. Placebo-controlled, double-blind trials reduce bias. [2].
12. (a) Withdrawal symptoms may occur [2]. (b) Dependence has altered normal nervous-system function and the drug is no longer present [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].
13. (a) $64 - 2 = 62$ [2]. (b) The values change from 2 to 64, with the highest value 64. [2]. (c) Antibiotics select existing resistant variants, which reproduce and increase in frequency [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].
14. (a) Substance concentration [1]. (b) Diameter of inhibition zone [1]. (c) Same bacterium, agar, disc size, volume and incubation [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use replicates, aseptic technique and a concentration series [1].
15. (a) a state in which stopping a drug causes withdrawal [2]. (b) Award linked points from: Alcohol slows nervous-system activity, impairs judgement and coordination, and long-term misuse can damage liver and brain. Heroin is a powerful depressant and analgesic that can cause tolerance, dependence and dangerous respiratory suppression. Addiction involves compulsive use despite harm. [3]. (c) Correct statement: placebo means an inactive treatment used as a comparison [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.